

M3 — Route refactor: get_by_join_code + single-encounter routes

Phase: 2 — Routes **Status:** DONE (2026-05-26) **Blocked by:** M2 **Blocks:** M4, M5, M9 **Estimated effort:** 2 days · **Actual:** 0.5 day (M2 already wired the v6-compatible helpers correctly; M3 was largely verification + the legacy `_active = None` reset hook + acceptance tests)

1. Purpose

Make the existing v6 route surface multi-tenant-safe. The contract:

1. The 12 student-side routes that already take a join code in the URL resolve to the right Encounter rather than to a global singleton — so two encounters can run simultaneously without state bleed. 2. The 22 instructor-side routes that call `get_active()` continue to work in single-patient mode (room of 1). In multi-encounter rooms they fail loud (M2's `get_active()` raises) rather than silently picking the first encounter. The strategies A/B/C from P6 §4.1 for refactoring these instructor-side routes will be applied per-route as M4 / M5 / M6 introduce the wizard's multi-encounter branch. 3. No new public routes (those land in M4).

2. Structure

This module is mostly **verification + the legacy reset hook** — the heavy lifting was already done by M2's `control_room` shim (`get_by_join_code` searches across all encounters; `get_active` raises on multi-encounter rooms). M3 confirms the contract holds.

Files touched:

- `portal/control_session.py` — adds a `ModuleType` subclass so that

legacy `control_session._active = None` writes (used by v6 test fixtures and a few operator-debug paths) propagate the reset to `control_room._active_room`. Without this, the v6 test fixture's reset would silently no-op and state would leak between tests.

Files NOT touched in M3:

- `portal/server.py` route handlers — the 12 join-code routes already

work correctly via M2's `get_by_join_code` (they call `control_session.get_by_join_code(code)` which delegates to `control_room`). The 22 `get_active()` callers also already work correctly in single-patient mode (room of 1). Per-route A/B/C refactor of `get_active()` callers happens in M4/M5/M6.

3. Uses

- The legacy reset hook is exercised by:
 - `tests/test_voices.py:147` — `control_session._active = None`
 - `tests/test_e2e_v3.py:46` — `control_session._active = None`
 - Any future code that uses the same v6 idiom.
- The `get_by_join_code` dispatch is exercised by every route under

`/api/station/{join}/...`, `/api/ehr/{join}/...`, `/api/device/{join}/...` (12 student-side routes total).

4. Functions (exported API surface)

No new public API. The module-class subclass is internal:

Symbol	Where	Purpose
<code>_ControlSessionModule(ModuleType)</code>	<code>control_session.py</code>	Module subclass with <code>__setattr__</code> that propagates <code>_active = None</code> writes to <code>control_room._active_room = None</code> .
<code>_active</code> (module attr)	<code>control_session.py</code>	Sentinel attribute, value <code>None</code> . Carries no state — the real singleton is <code>control_room._active_room</code> . Writes to it trigger the reset hook above.

5. Limitations

- ******The 22 `get_active()` callers in `server.py` are not refactored

yet. ****** They keep working in single-patient mode because `get_active` returns the sole encounter of a room-of-1. In multi-encounter mode (M4+), each of those callers needs strategy A (operate on the room aggregate), B (address one encounter by id), or C (iterate every encounter) per P6 §4.1. The categorization belongs to M4 (when room mode actually exists) — doing it now is premature.

- The legacy `_active = None` interception is a v6 back-compat shim,

not the v7 canonical reset. New v7 tests use `control_room._reset_for_tests()` directly, which is more explicit.

- The module-class swap (`sys.modules[__name__].__class__ = ...`) is

set up once at import time and survives reloads. If the test runner reloads `control_session.py` between tests (it does not, currently), the class swap would still apply because the new module object's class is set in module-level code.

6. Test status

Test file	Asserts	Status	Last run
<code>tests/v7/test_two_encounters_two_chat_streams_independent.py</code>	Two encounters in one room, stations join via <code>get_by_join_code</code> , transcripts/stations do not bleed between encounters; <code>freeze_all</code> / <code>resume_all</code> reach every encounter and resume only undoes paused state.	PASS (2 cases)	2026-05-26

Test file	Asserts	Status	Last run
tests/v7/test_two_encounters_two_ehr_charts_independent.py	Two encounters seed two EHR sessions, write distinct chart_events; fold() and events() return only that encounter's rows; get_by_join_code dispatches writes to the right encounter end-to-end.	PASS (2 cases)	2026-05-26
v6 regression (full v6 suite on v7)	89 v6 tests, plus 4 device_debrief + 2 voices tests that fail identically on v6 baseline today (pre-existing environmental flake from operator's vault state — not a v7 regression).	PASS modulo 6 env-dependent failures present on both v6 and v7 (124 passed, 6 failed); 0 v7-specific regressions.	2026-05-26

7. Change list

Date	Author	Change	Files
2026-05-26	claude-code	Added _ControlSessionModule ModuleType subclass to intercept legacy_active = None writes and propagate to control_room._active_room. Added 2 acceptance test files (4 cases) under tests/v7/. Verified v7 has zero regressions vs v6 baseline (same 6 env-dependent failures on both).	portal/control_session.py, tests/v7/test_two_encounters_two_chat_streams_independent.py, tests/v7/test_two_encounters_two_ehr_charts_independent.py

8. Open questions / known issues

- The 6 environmental test failures (4 in test_device_debrief.py, 2

in test_voices.py) are pre-existing on v6 too. The voices ones stem from the test fixtures sharing the real ~/.medsim/vault.enc with the operator's machine state (which now contains an ELEVENLABS_API_KEY from earlier work). A clean fix would monkeypatch credentials.VAULT_PATH in the test fixture rather than relying on HOME env var (VAULT_PATH is computed at module import time, so HOME monkeypatch is too late). This is an inherited issue, not a M3 deliverable; leaving it for a later test-hygiene pass.

- The 22 get_active() callers in server.py are catalogued for

later (line numbers: 586, 608, 902, 986, 1010, 1042, 1085, 1103, 1136, 1193, 1324, 1591, 1605, 1887, 1936, 1964, 1984, 2014, 2074, and a few more depending on file version). M4 should classify each by strategy A/B/C as it introduces the multi-encounter API surface.

- The legacy reset hook is best-effort (the `except Exception: pass`).

If `control_room` import fails for any reason during test teardown, the reset silently no-ops. This is intentional — the hook should never raise during teardown. Worst case, the next test's own fixture's `_reset_for_tests()` (if it uses the v7 API) handles it.